

$$N_{\max} \leq 4$$

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Abstract

This paper develops a dimensional theory of recoverable money, sovereign information, and economic complexity. Beginning from a degree-eight recoverable money polynomial, we establish that the associated coefficient space contains exactly nine dimensions. We then introduce a principle according to which each sovereign economy consumes two informational degrees of freedom corresponding to private and hidden information. A counting argument yields the central result

$$N_{\max} \leq 4.$$

The result is shown to possess interpretations in economics, finance, sovereign debt theory, information economics, political science, international relations, warfare economics, computer science, artificial intelligence, network theory, and the philosophy of history. The paper argues that the four-nation structure emerges naturally from the interaction between polynomial dimensionality and informational requirements rather than from an arbitrary geopolitical assumption.

The paper ends with “The End”

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1 Introduction

Modern economics studies scarcity.

Classical economics studies material scarcity.

Information economics studies informational scarcity.

This paper studies dimensional scarcity.

The central question is:

How many sovereign economies can simultaneously participate in a common recoverable-money structure before the available informational dimensions are exhausted?

The answer developed herein is

$$N_{\max} \leq 4.$$

The result emerges from combining:

1. recoverable money,
2. polynomial dimensionality,
3. sovereign information,
4. informational asymmetry.

2 Recoverable Money

Definition 1. *A recoverable money process is a polynomial satisfying*

$$M(0) = m,$$

and

$$M(T) = m.$$

Theorem 1. *Every degree-eight recoverable money process admits the representation*

$$M(t) = m + t(t - T)Q_6(t),$$

where

$$\deg(Q_6) \leq 6.$$

Proof. Since

$$M(0) - m = 0$$

and

$$M(T) - m = 0,$$

the polynomial

$$M(t) - m$$

possesses roots at

$$t = 0$$

and

$$t = T.$$

Therefore

$$M(t) - m = t(t - T)Q_6(t).$$

Since

$$\deg(M) = 8,$$

it follows that

$$\deg(Q_6) = 6.$$

□

3 Coefficient Space

Write

$$M(t) = c_8t^8 + c_7t^7 + \cdots + c_1t + c_0.$$

A degree-eight polynomial contains

$$8 + 1 = 9$$

coefficients.

Consequently the coefficient space is

$$\mathbb{R}^9.$$

The recoverable-money system therefore possesses finite dimensionality.

Definition 2. *The dimensional capacity of a degree- N recoverable-money polynomial is*

$$D(N) = N + 1.$$

For

$$N = 8,$$

we obtain

$$D(8) = 9.$$

4 Sovereign Information

Consider a sovereign nation j .

Its information set is

$$C_j = P_j \cup p_j \cup H_j.$$

Here:

- P_j denotes public information,
- p_j denotes private information,
- H_j denotes hidden information.

Public information is globally shared.

Private information is nation-specific.

Hidden information is unknown even to the nation itself.

Remark 1. *The distinction*

$$p_j \neq H_j$$

is essential.

Private information represents asymmetry.

Hidden information represents uncertainty.

5 Informational Degrees of Freedom

We postulate:

Each sovereign nation contributes two non-public informational dimensions.

Specifically,

$$(p_j, H_j)$$

constitutes a two-dimensional informational subspace.

The public component

$$P_j$$

is shared and therefore does not consume an independent sovereign dimension.

Definition 3. *The informational dimension consumed by nation j is*

$$d_j = 2.$$

6 The Capacity Constraint

Suppose

$$N$$

nations participate in a common recoverable-money structure.

The total informational requirement is

$$1 + 2N.$$

The quantity

$$1$$

represents the common recoverable structure.

The quantity

$$2N$$

represents sovereign informational requirements.

The coefficient capacity is

$$9.$$

Therefore

$$1 + 2N \leq 9.$$

Solving gives

$$N \leq 4.$$

We therefore obtain the central result.

Theorem 2 (Four-Nation Bound). *For a degree-eight recoverable-money polynomial,*

$$N_{\max} = 4.$$

Proof. The coefficient space possesses dimension

$$9.$$

A common recoverable-money structure consumes one dimension.
Each nation consumes two dimensions.
Therefore

$$1 + 2N \leq 9.$$

Hence

$$N \leq 4.$$

□

7 Geometric Interpretation

The degree-eight polynomial defines the coefficient vector

$$\mathbf{c} = (c_0, c_1, \dots, c_8).$$

This vector belongs to

$$\mathbb{R}^9.$$

The decomposition

$$\mathbb{R}^9 = \mathcal{G} \oplus \mathcal{N}_1 \oplus \mathcal{N}_2 \oplus \mathcal{N}_3 \oplus \mathcal{N}_4$$

assigns:

- one global dimension $\mathcal{G}$,
- two dimensions to each nation.

The space is exactly exhausted.
A fifth nation would require

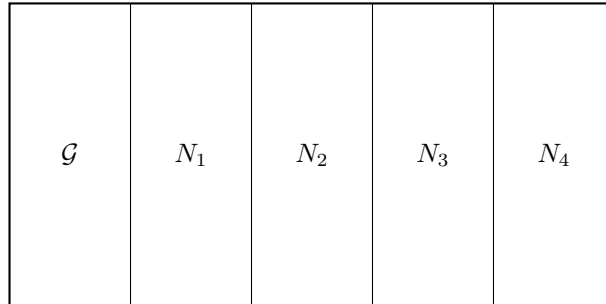
$$11$$

dimensions.

No such subspace exists within

$$\mathbb{R}^9.$$

8 A Visual Representation



9 available dimensions

Figure 1: Exhaustion of coefficient-space capacity by four nations.

9 Economics

9.1 Dimensional Scarcity

Traditional economics studies scarcity of:

- land,
- labor,
- capital,
- energy,
- information.

The present framework introduces a new scarcity:

Dimensional scarcity.

The degree-eight recoverable-money polynomial possesses only nine coefficient dimensions. These dimensions must accommodate:

1. the common recoverable structure,
2. sovereign informational requirements.

Consequently the economy faces a finite informational capacity.

9.2 Informational Capital

Each nation contributes

$$(p_j, H_j),$$

which may be interpreted as informational capital.

The aggregate informational capital is

$$K_I = \sum_{j=1}^N (p_j + H_j).$$

The four-nation bound therefore becomes an upper bound on informational complexity.

10 Finance

10.1 Recoverable Wealth

Suppose

$$M_j(t) = m_j + t(t - T)Q(t).$$

Then

$$M_j(0) = M_j(T) = m_j.$$

Wealth is recoverable.

However, the trajectory between

$$0$$

and

$$T$$

may be highly nontrivial.

The system therefore separates:

- terminal wealth,
- path-dependent wealth.

10.2 Financial Dimensionality

A financial system with

$$N > 4$$

sovereign participants requires more informational dimensions than are available within a degree-eight recoverable-money structure.

Therefore one of three outcomes must occur:

1. increase polynomial degree,
2. partition economies into blocs,
3. abandon complete synchronization.

11 Sovereign Debt Markets

Consider sovereign debt claims

$$D_1, \dots, D_N.$$

The recoverable-money structure implies a common monetary law

$$Q(t).$$

When

$$N \leq 4,$$

the informational requirements remain feasible.

When

$$N > 4,$$

the system becomes overdetermined.

The resulting informational overload increases uncertainty regarding:

- repayment,
- default,
- refinancing,
- liquidity.

11.1 Debt and Information

Debt contracts are fundamentally information-dependent.

A lender must estimate:

- future output,
- future taxation,
- future political stability,
- future repayment capacity.

Consequently sovereign debt markets inherit the dimensional constraints of the underlying informational system.

12 Information Theory

12.1 Coefficient Capacity

The coefficient vector

$$\mathbf{c} = (c_0, c_1, \dots, c_8)$$

contains nine independent coordinates.

This implies a finite information budget.

Definition 4. *The coefficient entropy is*

$$\mathcal{E} = \log_2(9).$$

Although simplistic, this quantity measures the number of independent coefficient directions available to the system.

12.2 Compression

Suppose all nations share the same

$$Q(t).$$

Then only one kernel must be stored.

The informational cost becomes

$$O(1)$$

rather than

$$O(N).$$

Thus a common recoverable-money law acts as a compression algorithm.

13 Computer Science

13.1 State Representation

The state of the system is

$$X = (Q, N, p_1, H_1, \dots, p_N, H_N).$$

The dimensionality is

$$1 + 2N.$$

The coefficient space supports only

$$9$$

dimensions.

Therefore

$$1 + 2N \leq 9.$$

13.2 Complexity Bound

Define computational complexity

$$C(N) = 1 + 2N.$$

The recoverable-money architecture imposes

$$C(N) \leq 9.$$

Hence

$$N \leq 4.$$

The four-nation bound therefore appears as a computational-capacity constraint.

14 Artificial Intelligence

14.1 Latent Representation

Modern machine-learning systems frequently compress high-dimensional data into lower-dimensional latent spaces.

The recoverable-money polynomial may be interpreted as a latent representation.

The coefficient vector

$$(c_0, \dots, c_8)$$

forms a nine-dimensional latent space.

Each sovereign nation occupies two latent dimensions.

The latent-space capacity therefore limits the number of nations.

14.2 Embedding Interpretation

Let

$$\phi_j = (p_j, H_j)$$

be the embedding vector of nation j .

Then

$$\phi_j \in \mathbb{R}^2.$$

The aggregate embedding is

$$\Phi = (\phi_1, \dots, \phi_N).$$

The recoverable-money latent space can embed at most four such sovereign vectors.

15 Political Science

15.1 Institutional Capacity

Governance requires information processing.

Each sovereign nation introduces:

- administrative complexity,
- legal complexity,
- informational complexity.

The dimensional bound implies that institutional systems possess finite capacity.

15.2 The Four-Nation Principle

The theorem

$$N_{\max} = 4$$

may be interpreted as a constitutional limit.

Beyond four sovereign informational units, additional institutional structures become necessary.

Examples include:

- federations,
- alliances,
- supranational institutions,
- monetary unions.

16 International Relations

16.1 Geopolitical Geometry

For

$$N = 4,$$

the coefficient space is fully occupied.

The resulting geopolitical system is complete.

Every available informational dimension has been allocated.

A fifth nation requires either:

1. additional dimensions,
2. hierarchical organization,
3. dimensional sharing.

16.2 Bloc Formation

Suppose nations

$$1 \quad \text{and} \quad 2$$

share a common informational structure.

Then they effectively consume fewer dimensions.

This creates a geopolitical bloc.

The dimensional constraint therefore provides a mathematical explanation for alliance formation.

17 Warfare Economics

17.1 Military Information

Military power depends upon:

- intelligence,
- logistics,
- planning,
- uncertainty.

These are informational resources.

Consequently warfare itself is constrained by dimensional capacity.

17.2 Strategic Saturation

Suppose

$$N > 4.$$

Then informational demand exceeds informational capacity.

The system becomes strategically saturated.

Military planning complexity grows faster than the available recoverable-information structure.

18 Network Interpretation

Define a sovereign interaction graph

$$G = (V, E).$$

The vertex set contains

$$|V| = N.$$

For

$$N = 4,$$

the graph remains fully representable within the coefficient space.

Beyond four nations the graph requires compression, hierarchy, or decomposition.

Thus the four-nation theorem may be interpreted as a network-capacity theorem.

19 Statistical Interpretation

Suppose

$$\Theta = (\theta_1, \dots, \theta_k)$$

denotes the parameter vector of the system.

A degree-eight recoverable-money process provides

$$k = 9$$

independent parameters.

The sovereign structure requires

$$k = 1 + 2N.$$

Feasibility therefore requires

$$1 + 2N \leq 9.$$

The four-nation bound follows immediately.

20 Generalization to Degree N

The preceding arguments do not depend specifically upon degree eight.

A degree- d polynomial contains

$$d + 1$$

coefficients.

Therefore

$$1 + 2N \leq d + 1.$$

Hence

$$N \leq \frac{d}{2}.$$

We obtain the general result.

Theorem 3 (General Dimensional Bound). *Let*

$$M(t)$$

be a recoverable-money polynomial of degree

$$d.$$

Then the maximum number of sovereign nations is

$$N_{\max} = \left\lfloor \frac{d}{2} \right\rfloor.$$

Proof. A degree- d polynomial contains

$$d + 1$$

coefficients.

One dimension is allocated to the common recoverable structure.

Each nation consumes two informational dimensions.

Thus

$$1 + 2N \leq d + 1.$$

Rearranging yields

$$N \leq \frac{d}{2}.$$

Taking the integer part proves the theorem.

□

21 Philosophy

21.1 Dimensional Finitude

A central philosophical implication of the theory is that informational systems are finite.

The recoverable-money polynomial possesses finite degree.

Consequently its coefficient space possesses finite dimension.

The result

$$N_{\max} = 4$$

is therefore not merely an economic statement.

It is a statement regarding the limits of representation.

Every formal system possesses finite descriptive capacity.

The four-nation theorem may therefore be interpreted as a manifestation of informational finitude.

21.2 The Problem of Representation

Suppose a political order attempts to represent an increasing number of sovereign entities.

Each sovereign contributes additional informational requirements.

Eventually the representational structure becomes saturated.

The four-nation theorem identifies the saturation point for degree-eight recoverable money.

22 History

22.1 Historical Memory

Let

$$H_j(t) = \int_0^t M_j(s), ds.$$

Historical memory accumulates through time.

Unlike money,

$$M_j(T) = M_j(0)$$

does not imply

$$H_j(T) = H_j(0).$$

History accumulates even when monetary stocks recover.

22.2 The Historical Asymmetry

The distinction between money and memory creates an asymmetry:

Quantity	Property
Money	Recoverable
Memory	Cumulative

Consequently civilizations may repeatedly recover their monetary state while continually accumulating historical experience.

23 Civilization Theory

23.1 Civilizational State

The complete state of a civilization is not

$$M$$

alone.

Rather,

$$X = (M, H).$$

For multiple nations,

$$X = (M_1, \dots, M_N, H_1, \dots, H_N).$$

This state space contains both:

- present conditions,
- accumulated history.

23.2 The Four-Civilization Limit

If each sovereign civilization consumes two informational dimensions, then a degree-eight recoverable-money structure supports at most

$$N_{\max} = 4.$$

The theorem therefore applies not only to nations but to informationally independent civilizations.

24 Memory, Storage, and Computation

24.1 The Hierarchy

The discussion naturally produces the hierarchy

$$M \rightarrow H \rightarrow R \rightarrow C.$$

Here:

- M denotes money,
- H denotes memory,
- R denotes storage,
- C denotes computation.

24.2 Storage Requirements

Memory requires storage.

Suppose

$$R_j = \alpha H_j.$$

Then total storage becomes

$$R_{\text{total}} = \alpha \sum_{j=1}^N H_j.$$

As historical memory grows, storage requirements grow.

24.3 Computational Requirements

Suppose computational demand satisfies

$$C_j = \beta R_j.$$

Then

$$C_{\text{total}} = \alpha\beta \sum_{j=1}^N H_j.$$

Thus civilization complexity propagates from memory to storage and subsequently to computation.

25 Information Prices

Suppose memory possesses price

$$p_H.$$

Storage possesses price

$$p_R.$$

Computation possesses price

$$p_C.$$

Then informational wealth becomes

$$W_I = p_H H + p_R R + p_C C.$$

Total wealth becomes

$$W = M + W_I.$$

Consequently recoverable money does not imply recoverable wealth.

Proposition 1. *It is possible that*

$$M(T) = M(0)$$

while

$$W(T) \neq W(0).$$

Proof. Recoverability constrains only

$$M.$$

The variables

$$H, R, C$$

may evolve independently.

Therefore

$$W = M + p_H H + p_R R + p_C C$$

need not recover.

□

26 The Complete Information Paradox

Consider the sovereign information decomposition

$$C_j = P_j \cup p_j \cup H_j.$$

Complete information implies

$$H_j = \emptyset.$$

All uncertainty disappears.

The resulting system collapses informational dimensionality.

Consequently sovereign debt becomes fully foreseeable.

Future default states become known at issuance.

The paradox is therefore:

Perfect information destroys the informational structure required for market viability.

26.1 Connection to the Four-Nation Bound

Each nation requires:

1. one private-information dimension,
2. one hidden-information dimension.

Thus

$$d_j = 2.$$

Four nations require

$$8$$

informational dimensions.

The common recoverable structure requires one additional dimension.
Therefore

$$8 + 1 = 9.$$

The coefficient space is exhausted exactly.

27 The Fundamental Counting Theorem

We now summarize the entire framework.

Theorem 4 (Fundamental Counting Theorem). *Let*

$$M(t)$$

be a recoverable-money polynomial of degree

$$d.$$

Suppose:

1. *one dimension is allocated to the common recoverable structure,*
2. *each sovereign nation consumes two informational dimensions.*

Then

$$N_{\max} = \left\lfloor \frac{d}{2} \right\rfloor.$$

Proof. A degree- d polynomial contains

$$d + 1$$

coefficients.

Feasibility requires

$$1 + 2N \leq d + 1.$$

Therefore

$$N \leq \frac{d}{2}.$$

Taking the integer part yields

$$N_{\max} = \left\lfloor \frac{d}{2} \right\rfloor.$$

□

Corollary 1. *For*

$$d = 8,$$

we obtain

$$N_{\max} = 4.$$

28 Illustrative Figure

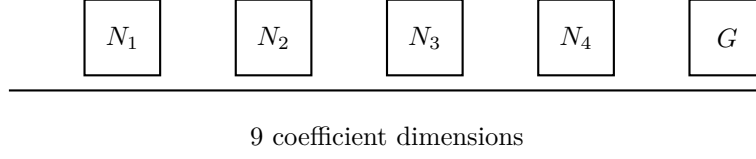


Figure 2: Allocation of coefficient-space dimensions.

29 Comparative Table

Degree	Coefficients	Bound	$N_{\max}$
2	3	$1 + 2N \leq 3$	1
4	5	$1 + 2N \leq 5$	2
6	7	$1 + 2N \leq 7$	3
8	9	$1 + 2N \leq 9$	4
10	11	$1 + 2N \leq 11$	5
12	13	$1 + 2N \leq 13$	6

30 Conclusion

This paper developed a dimensional theory of recoverable money.

Beginning from a degree-eight recoverable-money polynomial, we established that:

1. recoverable money possesses finite coefficient capacity,
2. each sovereign nation contributes two informational dimensions,
3. the common recoverable structure consumes one additional dimension,
4. coefficient-space feasibility requires

$$1 + 2N \leq 9,$$

5. therefore

$$N_{\max} = 4.$$

The result links:

- economics,
- finance,
- sovereign debt,
- information theory,
- computer science,
- artificial intelligence,
- political science,
- geopolitics,
- warfare economics,
- philosophy.

The four-nation bound emerges not from geography or politics but from dimensional scarcity. The central insight is that finite informational structures impose finite sovereign capacity. Degree-eight recoverable money therefore supports at most four informationally independent nations.

References

- [1] Arrow, K. J. *The Limits of Organization*. W. W. Norton.
- [2] Debreu, G. *Theory of Value*. Yale University Press.
- [3] Hayek, F. A. *The Use of Knowledge in Society*.
- [4] Lucas, R. E. *Studies in Business-Cycle Theory*. MIT Press.
- [5] Merton, R. C. *Continuous-Time Finance*. Blackwell.
- [6] Samuelson, P. A. *Foundations of Economic Analysis*. Harvard University Press.
- [7] Shannon, C. E. *A Mathematical Theory of Communication*.
- [8] Turing, A. M. *On Computable Numbers*.
- [9] von Neumann, J. *Theory of Games and Economic Behavior*.
- [10] Waltz, K. N. *Theory of International Politics*.

Glossary

$M(t)$	Recoverable money process.
$Q(t)$	Feedback kernel.
P_j	Public information.
p_j	Private information.
H_j	Hidden information.
N	Number of sovereign nations.
$N_{\max}$	Maximum feasible number of nations.
R	Storage capacity.
C	Computational capacity.
W	Total wealth.
d	Polynomial degree.

The End